

Readable but Not Steerable: Evidence Sufficiency and Uptake in the Residual Stream of Retrieval-Augmented LLMs

Anonymous

Abstract

When retrieved evidence becomes sufficient to answer a question, does a language model undergo an internal state transition that predicts whether it will use that evidence, before it generates a token? We study this with paired forward passes over semantically defined *evidence trajectories* for multi-hop questions: for each question we present nested contexts that add distractors, individual reasoning hops, the final decisive hop, redundant support, falsified evidence, or the answer string in an irrelevant passage, and we read the residual stream at the answer-start position. Linear probes on the paired activation update $\Delta h = h(C_k) - h(C_{k-1})$ recover *sufficiency* (the increment that completes the evidence) at 0.95 AUROC on document-, answer-entity- and relation-disjoint splits, transfer without retraining to a conversational protocol, to a second dataset and to deeper questions, and replicate across three model families. A fine-tuned text classifier, however, recovers sufficiency almost as well. A second, orthogonal direction predicts *uptake*, whether decisive evidence will actually be integrated rather than ignored, at 0.78 AUROC; here text models stay near chance (0.56), at full budget 0.63, so uptake is a genuinely internal signal. Causal tests then separate readout from mechanism. The discriminative probe directions are inert under steering and projection-out in every model, whereas patching the full anchor residual is causally sufficient and its effect lies entirely in the orthogonal complement of the probe direction, in a high-rank subspace that on some models is dominated by the answer’s unembedding direction. Only the mean-difference sufficiency direction is causally active, and it behaves as an abstention gate: projecting it out silences answering, and steering along it raises answering on distractors and false evidence as much as on sufficient evidence. High-AUROC, transferable probes thus find the label, not the lever, and the one lever we find controls whether the model answers, not whether it has accommodated the evidence.

1 Introduction

Retrieval-augmented generation supplies a language model with passages and asks it to answer from them. Two internal questions matter for such systems: has the model registered that the supplied evidence is now *sufficient*, and has it *integrated* the decisive passage into its answer computation? A reliable pre-generation readout of either would let a system decide whether to retrieve again, detect ignored evidence, or abstain; a causal handle on either would let a system correct the model’s behaviour directly. Prior work has shown that residual-stream activations carry signals about context-versus-parametric conflict and context reliance, but these are typically identified on single contexts, and it is rarely tested whether the directions that read such states also control them.

We take a different unit of analysis: the *paired update*. For each multi-hop question we construct nested evidence contexts $C_0 \subset C_1 \subset \dots$ and run a fresh, stateless forward pass on each, reading the residual stream at the fixed answer-start position. The difference $\Delta h_k = h(C_k) - h(C_{k-1})$ isolates what one added paragraph did to the model’s state. Because every decisive increment (the paragraph completing the evidence) is matched with controls that add a length-matched distractor, a duplicate of already-present support, a falsified version of the decisive paragraph, or the answer string embedded in an irrelevant passage from the same penultimate state, a probe trained on these updates cannot succeed by detecting length, answer-string presence or passage relevance alone. Each increment is also labelled with the model’s own behavioural transition (wrong→correct, wrong→wrong, correct→wrong, and so on), giving separate supervision for sufficiency, uptake, correction and stability rather than one conflated “answer-update” target.

Our contributions are as follows.

- **A matched-increment protocol** for reading evidence-sufficiency and evidence-uptake states from single forward passes, with four-way disjoint evaluation (question, document, answer entity, relation type) and a conversational-revision ablation (Sections 2 and 3).
- **Readout results.** Sufficiency is linearly decodable at 0.95 AUROC in Qwen2.5-7B, Llama-3.1-8B and Mistral-7B, survives residualising twelve nuisance covariates, orders increment kinds as predicted (decisive > false > distractor > redundant > answer-string control), and transfers without retraining to a conversational protocol (0.79), to 2WikiMultiHopQA (0.96) and to 3–4-hop questions (0.82). Uptake is decodable at 0.78 along a direction nearly orthogonal to sufficiency (Section 4).
- **Text baselines.** A fine-tuned DeBERTa that sees the full context recovers sufficiency nearly as well as the probe, but not uptake (0.56 vs 0.78): uptake is the internal signal (Section 4.1).
- **Causal dissociation.** Discriminative probe directions are inert under additive steering (up to 35% of the residual norm), projection-out and component patching, in all three models. The full anchor residual is causally sufficient, and its effect lives in the orthogonal complement, a high-rank subspace whose composition is model-specific. The mean-difference sufficiency direction is causally active but acts as an abstention gate rather than an accommodation mechanism (Section 5).

The picture that emerges is that evidence sufficiency and uptake are cleanly *readable* before generation, that the readout of uptake is not recoverable from text, and that neither readout is a *lever*: the linear probe finds a correlate of the state, while the mechanism that turns evidence into an answer is high-dimensional and content-bearing.

2 Experimental setup

Data. MuSiQue [2] provides 2–4-hop questions with gold supporting paragraphs, per-hop decompositions with sub-answers and relation types, and distractors; our main runs use its two-hop training questions, a second run 3–4-hop questions, and 2WikiMultiHopQA [1] is the transfer set. For each model we screen a pool of 4,000 questions closed-book and keep 2,000 that the model answers incorrectly without evidence (closed-book accuracy is 1.4% for Qwen, 4.0% for Llama, 1.3% for 3–4-hop questions), so every increment concerns evidence the model actually needs. We keep the whole trajectory for every question, including stages after the answer first becomes correct, to avoid conditioning the data on the model’s behaviour.

Models. Qwen2.5-7B-Instruct (28 layers, $d = 3584$), Llama-3.1-8B-Instruct (32 layers, $d = 4096$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), all in bfloat16 with greedy decoding of at most 16 tokens. Prompts use each model’s chat template.

Prompt and anchor. Every stateless stage uses the same template: an instruction (“Use only the supplied evidence to answer the question. If the evidence is insufficient, output INSUFFICIENT.”), the evidence block, the question and an answer cue. We read the residual stream after every decoder block at the last prompt token, the position that emits the first answer token, so all readouts are pre-generation. An *evidence-augmented* regime (“use the evidence together with your own knowledge”) is run in parallel and gives the same results (Appendix C); we report the evidence-only regime unless stated.

Evidence conditions. For a question with gold hop paragraphs $g_1, \dots, g_n$ we build (Appendix A): no evidence C_0 ; matched distractors only C_D ; prefix chains $C_{o_1} \subset C_{o_1 o_2} \subset \dots$ in the canonical hop order, the reverse order and, for $n \geq 3$, random permutations, the last prefix being *minimally sufficient*; leave-one-out contexts for $n \geq 3$; the sufficient context plus a distractor, plus a duplicate support, or plus a conflicting falsified paragraph; and the full 20-paragraph context. New paragraphs are inserted at a random position of the previous presentation order, so increments are textually minimal while the position of the decisive paragraph is randomised. Falsified paragraphs replace the hop’s sub-answer with the sub-answer of another question of the same relation type; answer-string controls append a Wikipedia-style “not to be confused with {answer}” hatnote to a distractor.

Increments and labels. From every penultimate state S (all hops but one) we form the decisive increment $S \rightarrow S + g_{\text{last}}$ and four matched controls from the same S : distractor, redundant, false and answer-string. Together with first-hop, post-sufficiency and full-context increments this yields ≈ 17 conditions and 17 increments per two-hop question (35,008 conditions and 34,000 increments per model; 65,375 and 65,599 for 3–4 hops). Each increment carries the label $y^{\text{suff}} = \mathbf{1}[C_k \text{ sufficient}, C_{k-1} \text{ not}]$ and the model’s behavioural transition between the two stages, from which we define *uptake* (decisive increments: answer becomes correct vs. does not), *correction* (from a wrong or abstaining state: becomes correct vs. not), *stability* (from a correct state: stays correct vs. destabilised) and *revision* (answer changed). Correctness uses exact match on any alias or containment within a short prediction.

Behaviour. All three models abstain on 94–97% of distractor-only contexts and answer 64–75% of minimally sufficient ones, with 36–47% correct; falsified final hops halve abstention, so the models do follow false context (Table 5, Appendix C).

Splits. Probes are trained on 70% of questions and evaluated on the rest under four groupings never shared between train and test: question id, gold document (connected components of questions sharing a supporting page), answer entity, and relation-type tuple. We report the document split by default and all four in Table 6 (Appendix C).

3 Method

Paired updates and probes. Let $h^{(l)}(C) \in \mathbb{R}^d$ be the residual stream after block l at the anchor for context C . For increment $C_{k-1} \rightarrow C_k$ we form $\Delta h^{(l)} = h^{(l)}(C_k) - h^{(l)}(C_{k-1})$ and, for each task and layer, fit an ℓ_2 -regularised logistic regression on standardised features with class balancing, selecting the layer by grouped three-fold cross-validation on training questions only. We also report the *difference-of-means* direction $v_{\text{dm}} \propto \mathbb{E}[\Delta h \mid y=1] - \mathbb{E}[\Delta h \mid y=0]$ and, for the raw state $h(C)$, *state probes* for sufficiency and correctness. Directions are unit vectors in raw activation space.

Nuisance analysis. A paired difference also encodes added tokens, context length, answer-string presence, passage relevance and the change in the output distribution. We build a nuisance vector z per increment (token counts, gold-answer log-probability change, next-token entropy and top-probability change, answer-string presence, TF-IDF relevance, insertion position, and a 16-dimensional TF-IDF-SVD embedding of the added text; Appendix B) and report (i) a nuisance-only classifier, (ii) the probe on residuals of a ridge regression of Δh on z , and (iii) the AUROC gain of adding out-of-fold probe scores to the nuisance model.

Text baselines. To ask whether the internal state is needed at all, we train text-only classifiers on the same increments and splits: TF-IDF logistic regression and a fine-tuned DeBERTa-v3-base cross-encoder, each on “question [SEP] added paragraph” alone and with the previous context appended.

Transfer. Learned directions are applied without retraining to (i) the *conversational revision* protocol, in which the model sees its own previous answer and only the new evidence (“Your previous answer was A_k . New evidence: E_{k+1} . Answer again; change your answer only if the total evidence warrants it.”), pairing $h_{\text{conv}}(C_k \rightarrow C_{k+1}) - h(C_k)$; (ii) 2WikiMultiHopQA; and (iii) 3–4-hop MuSiQue including leave-one-out completions. Model-family transfer is replication of the procedure, since hidden sizes differ.

Causal tests. Interventions act on held-out questions of the document split. *Steering* adds $\alpha u v$ from the anchor onward at layer l (or all positions), where u is the separation between the sufficient and insufficient state means along v on training data, so $\alpha = 1$ moves an activation by one sufficient-minus-insufficient gap; a random unit direction with the same ℓ_2 magnitude is the control. *Projection-out* sets the component along v to its insufficient-state mean at all layers from l onward. *Patching* runs the insufficient prompt C_{n-1} with its anchor residual replaced, at every layer from l onward, by h_{suff} (full), $h_{\text{ins}} + \text{proj}_v(\Delta)$ (parallel), $h_{\text{ins}} + \Delta - \text{proj}_v(\Delta)$ (orthogonal), the projection of Δ onto a random direction, onto the top- k principal

Task	Qwen2.5-7B			Llama-3.1-8B			Mistral-7B		
	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain
Sufficiency	0.948 [0.94,0.95]	0.87	0.11	0.952 [0.95,0.96]	0.86	0.12	0.949 [0.94,0.95]	0.87	0.11
Sufficiency (matched)	0.911 [0.90,0.92]	0.90	0.07	0.918 [0.91,0.93]	0.89	0.08	0.910 [0.90,0.92]	0.90	0.07
Uptake	0.775 [0.74,0.80]	0.66	0.14	0.783 [0.75,0.81]	0.58	0.21	0.771 [0.74,0.80]	0.59	0.18
Correction	0.730 [0.70,0.76]	0.81	0.01	0.817 [0.79,0.84]	0.83	0.06	0.792 [0.77,0.81]	0.85	0.03
Stability	0.750 [0.72,0.78]	0.92	-0.01	0.869 [0.85,0.89]	0.90	0.02	0.800 [0.77,0.83]	0.90	0.01
State sufficiency	0.992 [0.99,0.99]	—	—	0.994 [0.99,0.99]	—	—	0.996 [1.00,1.00]	—	—

Table 1: Held-out AUROC of the best-layer probe on paired activation updates (document-disjoint split; 95% question-level bootstrap intervals), with the AUROC of a classifier on nuisance covariates alone and the gain from adding the probe score to that classifier. State-sufficiency probes act on the raw activation.

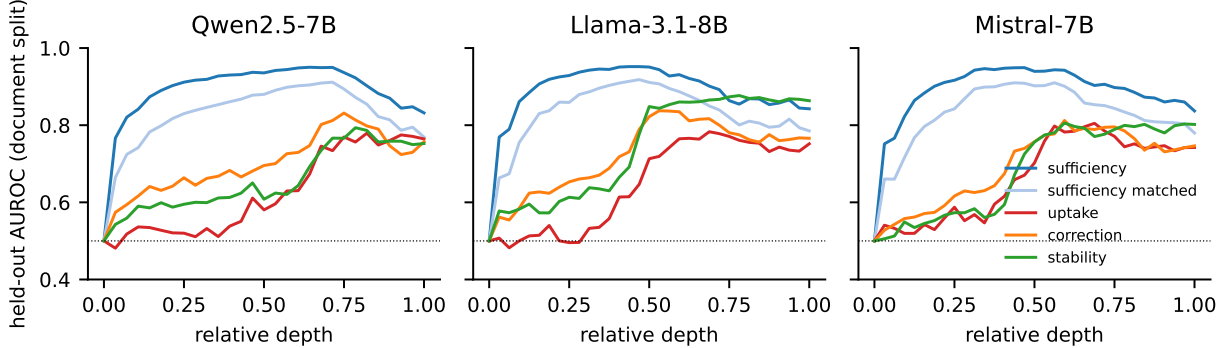


Figure 1: Held-out AUROC by relative depth for the update tasks (document split). Sufficiency peaks around mid-depth in every model; uptake and correction emerge later.

components of training deltas, or onto the gold answer’s first-token unembedding direction. Outcomes are the rate of non-abstention (“answered”) and of correct answers on 150 held-out questions per set; for false-evidence contexts we also report the rate of producing the injected false answer.

4 Readout results

Sufficiency is linearly readable and robust. Table 1 shows that the decisive increment is separated from all non-decisive increments at 0.948–0.952 AUROC in the three models, with the best layer at 50–60% depth (Fig. 1). Against the matched controls only (distractor, redundant, false, answer-string from the same penultimate state) the probe still reaches 0.91–0.92. The direction survives residualising the nuisance covariates (0.94–0.96) and adds +0.11 AUROC to a nuisance-only classifier. Table 6 shows the result is insensitive to the grouping: question-, document-, answer-entity- and relation-disjoint splits all give 0.94–0.96, and the state probe on the raw activation reaches 0.99. Projections of held-out increments on the sufficiency direction order the increment kinds exactly as hypothesised (Fig. 2): decisive > false > distractor > redundant > answer-string control > non-decisive gold, with post-sufficiency increments and the first hop far below. The false-evidence increment scoring second indicates that the direction encodes “the final slot has been filled” rather than truth, which is what the model can know before generation.

Uptake is a separate, later signal. Whether a decisive increment is actually integrated (answer becomes correct) versus behaviourally ignored is decodable at 0.77–0.78 AUROC (document split; 0.85 question split), at layers near the end of the network, and this is the task where the activation adds most beyond nuisance covariates (+0.14 to +0.21). The uptake direction is nearly orthogonal to the sufficiency direction (cosine 0.02–0.03) and the sufficiency direction reaches only 0.54–0.60 on the uptake task, while uptake and wrong-to-correct *correction* share a direction (cosine 0.30–0.34; cross-task AUROC 0.82; Fig. 3). Correction and stability, by contrast, are largely explained by the change in the output distribution: nuisance-only classifiers

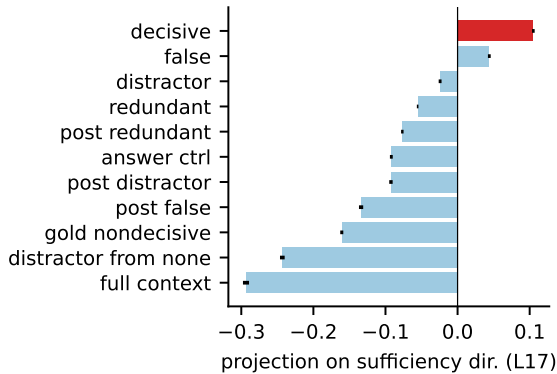


Figure 2: Mean projection of held-out increments on the sufficiency direction by increment kind (Qwen2.5-7B, document split; standard errors). Red: decisive.

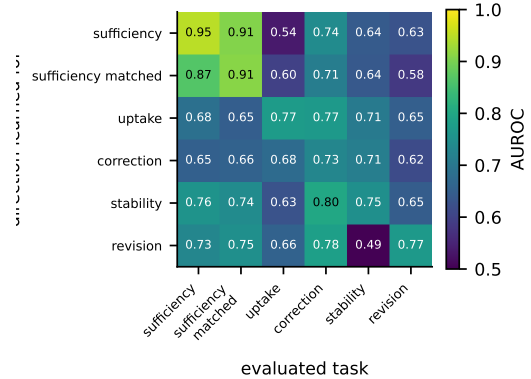


Figure 3: Cross-task transfer (Qwen2.5-7B, document split): AUROC of the direction learned for the row task on the column task.

Direction (MuSiQue 2-hop, stateless)	→ conversational	→ 2WikiMultiHopQA	→ MuSiQue 3–4 hop
sufficiency	0.790	0.956	0.822
sufficiency matched	0.776	0.931	0.849
uptake	0.776	0.791	0.675
correction	0.920	0.846	0.744
stability	0.820	0.822	0.755
revision	0.659	0.770	0.708

Table 2: Zero-retraining transfer of Qwen2.5-7B 2-hop directions (AUROC at the source’s best layer).

reach 0.81–0.85 and 0.90–0.92, and the probe adds at most 0.06.

Transfer. Table 2 applies the 2-hop stateless directions without retraining. To the conversational protocol they transfer at 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability), supporting the view that the transition is intrinsic to evidence integration rather than an artefact of being asked to reconsider. To 2WikiMultiHopQA the sufficiency direction transfers at 0.96 (0.93 matched), correction at 0.85 and uptake at 0.79; probes trained within 2WikiMultiHopQA reach 0.99, 0.89 and 0.91. To 3–4-hop questions sufficiency transfers at 0.82–0.85; within that run the leave-one-out completions score as high as chain completions (Appendix C), and sufficiency probes reach 0.93 on every split, but uptake falls to 0.61 on the document and answer splits (from 0.80 on the question and relation splits).

4.1 Is the internal state needed? Text-only baselines

Table 3 answers the obvious objection. For sufficiency, TF-IDF reaches 0.62 and a DeBERTa that sees only the question and the added paragraph 0.72–0.73, but a DeBERTa that also sees the previous context reaches 0.89–0.92 at a reduced training budget (one epoch, 256 tokens, 8k increments) and 0.97 at full budget (two epochs, 512 tokens, all increments), exceeding the probe’s 0.95 on the same split. Sufficiency is therefore largely recoverable from the text: the activation probe is an accurate readout of a property the input already determines. Uptake is the opposite case. On the same increments no text model exceeds 0.63 on whether the model will integrate the decisive paragraph (Qwen 0.56 reduced and 0.63 full budget, Llama 0.52, Mistral 0.60 with context), while the activation probes reach 0.77–0.78: whether evidence is taken up depends on the model’s internal state, not on the evidence alone.

Model	Budget	Task	TF-IDF	DeBERTa (q+added)	DeBERTa (+context)	Probe
Qwen2.5-7B	reduced budget	sufficiency	0.623	0.723	0.900	0.948
Qwen2.5-7B	reduced budget	sufficiency matched	0.623	0.758	0.823	0.911
Qwen2.5-7B	reduced budget	uptake	0.574	0.569	0.557	0.775
Qwen2.5-7B	full budget	sufficiency	0.611	0.723	0.972	0.948
Qwen2.5-7B	full budget	uptake	0.574	0.625	0.630	0.775
Llama-3.1-8B	reduced budget	sufficiency	0.625	0.715	0.888	0.952
Llama-3.1-8B	reduced budget	uptake	0.463	0.466	0.517	0.783
Mistral-7B	reduced budget	sufficiency	0.625	0.730	0.920	0.949
Mistral-7B	reduced budget	uptake	0.536	0.573	0.604	0.771

Table 3: Text-only classifiers on the same increments and splits versus the activation probe (document split).

Model	Patching the insufficient prompt (correct/answered)								Projection-out on sufficient (answered)		
	none	full	$\parallel v$	$\perp v$	rand-256	PC1	PC256	unemb.	v_{dm}	random	v_{lr}
Qwen2.5-7B	0.04/0.13	0.26/0.71	0.04/0.13	0.25/0.71	0.04/0.13	0.09/0.29	0.18/0.59	0.09/0.19	0.01	0.72	0.72
Llama-3.1-8B	0.02/0.11	0.24/0.74	0.02/0.12	0.24/0.73	0.02/0.12	0.02/0.19	0.09/0.41	0.19/0.62	0.00	0.73	0.75
Mistral-7B	0.07/0.17	0.09/0.60	0.07/0.17	0.09/0.60	0.07/0.17	0.08/0.61	0.08/0.56	0.07/0.19	0.01	0.63	0.63
Qwen 3-4 hop	0.05/0.20	0.25/0.62	0.06/0.19	0.25/0.62	0.06/0.20	0.07/0.23	0.20/0.51	0.07/0.21	0.03	0.63	0.63

Table 4: Causal tests on 150 held-out questions (document split), all layers from the best sufficiency layer onward. Left: the insufficient prompt C_{n-1} with its anchor residual patched (correct / answered). Right: the sufficient prompt with the component along a direction set to its insufficient-state value (answered). v_{dm} : difference-of-means sufficiency direction; v_{lr} : logistic-probe direction.

Closed-book transfer. Applied to closed-book activations for the 4,000 screened questions, the evidence-trained state-correctness direction predicts closed-book correctness at 0.81 AUROC (best layer 0.84) against a ceiling of 0.87 for a probe trained on closed-book data, and the abstention-flag direction predicts closed-book abstention at 0.98; only 1.5% of these questions are answered correctly closed-book, so we treat this as suggestive (Appendix C).

5 Causal results: readout versus mechanism

Probe directions are inert. Additive steering along the logistic sufficiency or uptake direction at the best layer changes no aggregate rate at any dose from 0.7 to 22 ℓ_2 units (35% of the residual norm, Appendix C), and a matched random direction is equally inert, so the null is not a matter of dose. Patching only the component of the sufficient-minus-insufficient update that lies along the probe direction, at every layer from the best one onward, leaves the insufficient prompt’s behaviour unchanged in all three models (Table 4, “ $\parallel v$ ”); projecting the logistic direction out of sufficient prompts changes nothing either (v_{lr} column). The same holds for the uptake direction.

The anchor residual is causally sufficient, in the orthogonal complement. Replacing the anchor residual of the insufficient prompt with the sufficient run’s, at all layers from the best layer onward, takes Qwen from 0.04/0.13 (correct/answered) to 0.26/0.71 and Llama from 0.02/0.11 to 0.24/0.74, roughly half the correctness the model achieves when the paragraph is actually present. Removing the probe component from that patch (“ $\perp v$ ”) leaves the effect intact (0.25/0.71, 0.24/0.73). The single anchor position therefore carries enough to make the model answer, and often answer correctly, but none of that content is along the direction the probe reads. Mistral shows the same dissociation with a weaker full-patch effect (0.09/0.60), as does the 3-4-hop run (0.25/0.62).

The causal content is high-rank and model-specific. Figure 4 patches the update projected on the top- k principal components of training updates. Answering rises gradually with rank (Qwen: 0.29 at $k=1$, 0.49 at $k=16$, 0.59 at $k=256$, 0.71 full) while a random 256-dimensional subspace does nothing; correctness

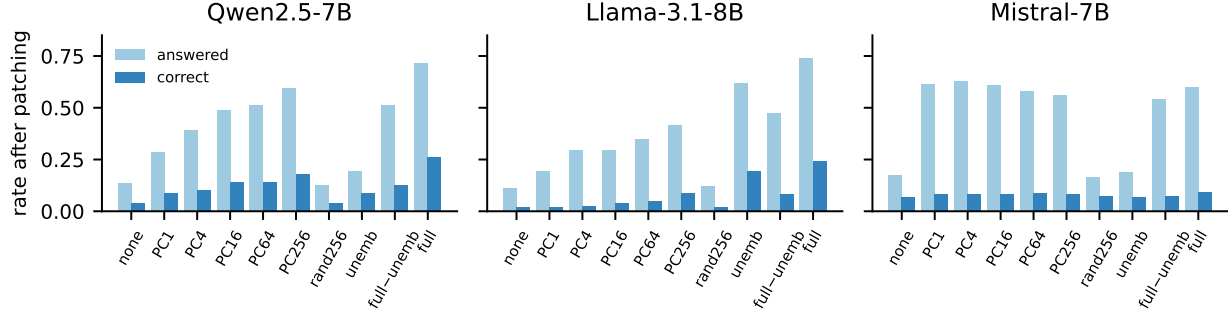


Figure 4: Rank of the causal subspace: the insufficient prompt patched with the sufficient update projected on the top- k principal components of training updates, on a random 256-dimensional subspace, on the gold answer’s unembedding direction (“unemb”), or on everything but that direction.

needs even more of the update. The first component carries much of the answer-versus-abstain switch, consistent with the projection-out result below. What carries correctness differs by model: on Llama the gold answer’s unembedding direction alone restores 0.19/0.62, most of the full effect, whereas on Qwen and Mistral it restores little (0.09/0.19, 0.07/0.19) although removing it from the full patch halves Qwen’s correctness (0.26 $\rightarrow$ 0.13). Mistral’s and the multi-hop run’s effects are captured by the top principal components instead (0.56 and 0.51 answered at $k=256$).

The mean-difference direction is a causal abstention gate. The one direction with a causal effect is v_{dm} , the mean difference between sufficient and insufficient states, which the discriminative probe does not align with. Setting its component to the insufficient-state value at all layers from the best one onward makes all three models abstain on essentially every sufficient prompt (answered 0.01, 0.00, 0.01 versus 0.72, 0.73, 0.63 for a random direction under identical treatment), and equally on false-evidence prompts. Low-dose additive steering along v_{dm} over four layers (Fig. 5; full numbers in Table 7, Appendix C) gives a graded, coherent, monotone dose-response in both directions with a flat random control: at $\alpha = -1$ Qwen answers 34% of sufficient prompts (from 71%) with unchanged precision among the answers it gives. The positive direction, however, is indiscriminate. At $\alpha = +1$ answering rises on distractor-only prompts (Qwen 0.03 $\rightarrow$ 0.12, Llama 0.03 $\rightarrow$ 0.29, Mistral 0.03 $\rightarrow$ 0.28), on insufficient prompts (0.13 $\rightarrow$ 0.26, 0.11 $\rightarrow$ 0.51, 0.17 $\rightarrow$ 0.47), on answer-string controls, and false-evidence following roughly doubles on Llama (0.19 $\rightarrow$ 0.39), while correct answers on sufficient prompts gain 3 points on Qwen and 18 on Llama. By the criterion we set in advance, a direction that makes the model accept any passage is a context-compliance or abstention direction, not an accommodation direction. Applied across all positions and all layers at $|\alpha| \geq 2$ the same recipe destroys the output for probe and random directions alike (Appendix C).

Adaptive retrieval. Stopping retrieval on the pre-generation state-sufficiency score (calibrated at 0.97 precision on training questions) reaches 0.44 accuracy in 2.9 rounds on held-out Qwen trajectories, versus 0.46 in 3.1 rounds for “stop when the model does not say INSUFFICIENT” and 0.475 for an oracle: the readout saves rounds but does not beat the model’s own abstention.

6 Discussion

Two signals, one lever, and none of them is accommodation. The results separate three things the plan conflates under “an internal state transition”. *Sufficiency* is a state of the input that the model represents cleanly and that a text model can also compute; it is a readout, and its causal counterpart is an abstention gate. *Uptake* is a state of the model: given the same decisive paragraph, whether the model will integrate it is predictable from its activations and not from the text, at 0.78 AUROC across three families. But neither readout direction is a lever, and the content that actually turns an insufficient state

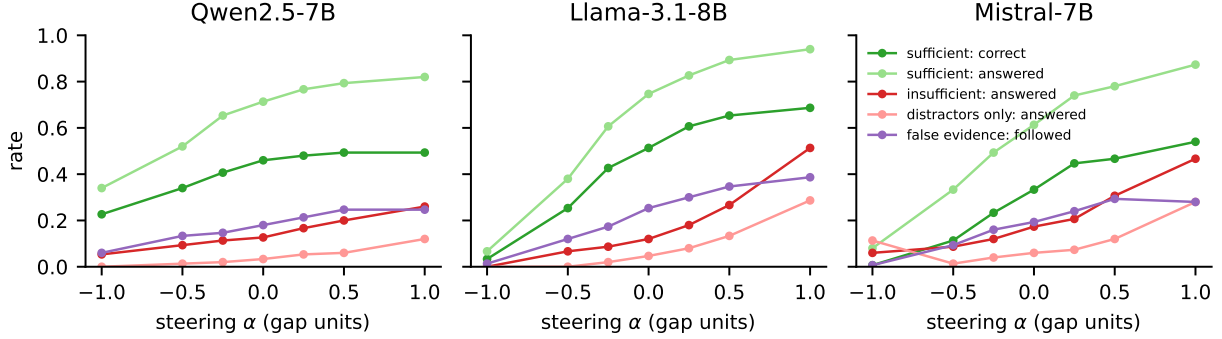


Figure 5: Dose-response of difference-of-means steering (four layers from the best sufficiency layer, anchor onward; α in units of the sufficient-minus-insufficient gap). Negative α induces abstention everywhere; positive α induces answering everywhere, including on distractors and false evidence.

into an answer is high-rank, orthogonal to both, and on some models concentrated on the answer’s own unembedding direction. Rather than a shared low-dimensional accommodation direction, the picture is a low-dimensional gate (answer or abstain) plus high-dimensional answer content.

Why probes do not find the lever. The logistic probe maximises separability, which it achieves along directions of low within-class variance; the mean-difference direction has a large component along a high-variance “answering” axis that the discriminative weight vector down-weights. The projection-out and rank experiments show the causal content is exactly in those high-variance directions. This is a concrete instance of the general warning that a high-AUROC, transferable probe direction can be causally irrelevant, and it argues for validating any proposed steering direction against the projection-out, rank and matched-random controls used here.

Practical reading. For retrieval systems the usable signal is uptake, as a detector of ignored evidence, rather than sufficiency, which the model’s own abstention already tracks about as well. The abstention gate is a genuine control knob, but it trades off abstention against compliance uniformly; it does not make the model more evidence-sensitive.

Limitations. Questions were selected to be closed-book failures, so the population is retrieval-dependent by construction. Redundant evidence is a verbatim duplicate. The uptake signal weakens on 3–4-hop questions under document- and answer-disjoint splits (0.61), and its test sets are small ($\approx 1,200$ increments per model). The full-budget text baseline was run for one model. Steering was tested with additive and projection interventions at the anchor and later positions; other intervention families (attention-head patching, path-specific interventions) may find levers we did not.

7 Related work

8 Conclusion

Paired activation updates over matched evidence trajectories give a clean experimental handle on what a language model registers when evidence arrives. They show that evidence sufficiency is readable before generation, transferable and text-recoverable; that evidence uptake is readable, transferable and *not* text-recoverable; and that neither readout direction is causally responsible for the behaviour it predicts. The one causal direction we find is an abstention gate. We release the trajectory construction, probing and intervention code to make the readout-versus-mechanism distinction a standard check for future work on evidence-aware generation.

References

- [1] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *COLING*, 2020.
- [2] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. In *Transactions of the Association for Computational Linguistics*, 2022.

A Evidence conditions and increments

For a question with n gold hop paragraphs $g_0, \dots, g_{n-1}$ (canonical order from the dataset’s decomposition) and its distractor pool, the conditions are:

- C_0 : no evidence; C_D : n distractors, each length-matched (word count) to one gold paragraph.
- Chains: for each hop order $o \in \{\text{canonical, reverse}\} \cup \{\text{random permutations, } n \geq 3\}$, the prefixes $C_{o_1}, C_{o_1 o_2}, \dots$; each new paragraph is inserted at a uniformly random position in the previous presentation order. The last prefix is minimally sufficient. Identical conditions arising from different orders are de-duplicated.
- Leave-one-out ($n \geq 3$): all gold paragraphs but one, shuffled, and its completion.
- From every penultimate state S (used by a chain or leave-one-out context): $S+D$ (distractor length-matched to the decisive paragraph), $S+R$ (verbatim duplicate of a gold paragraph already in S), $S+F$ (decisive paragraph with its sub-answer, and title if applicable, replaced by another question’s sub-answer of the same relation type; questions whose sub-answer does not appear in the paragraph have no false condition), $S+A$ (a distractor with “(Not to be confused with {answer}.)” appended).
- Post-sufficiency: the canonical minimal sufficient context plus a distractor, plus a duplicate gold, plus a conflicting falsified final hop; and the full context of all 20 paragraphs, shuffled.

Increment kinds and labels: `decisive/loo_decisive` ($y^{\text{suff}} = 1$), `gold_nondecisive`, `distractor`, `redundant`, `false`, `answer_ctrl`, `distractor_from_none`, `post_distractor`, `post_redundant`, `post_false`, `full_context`. The *sufficiency* task uses all kinds; the *matched* variant uses decisive increments and the four controls from the same penultimate state only. For 2-hop questions this gives 17 conditions and 17 increments per question; the conversational protocol re-runs the 10 chain-related increment kinds with the model’s own stateless answer as A_k (30,000 increments for Qwen).

B Probing, nuisance and intervention details

Activation capture. Activations are read by forward hooks on each decoder block’s output at the anchor position, in a single left-padded batched prefill that also appends the gold answer tokens after the anchor (causality leaves the anchor unaffected) to obtain the teacher-forced gold log-probability; the KV cache is then cropped and greedy decoding continues. Layer index l denotes the residual stream after block l ($l = 0$: embeddings).

Probes. `scikit-learn` logistic regression, $C = 0.5$, balanced class weights, features standardised on the training split; the layer is selected by grouped 3-fold cross-validation on training questions (never on test), and the reported number is the test AUROC at that layer. Directions are w/σ normalised to unit norm in raw space. The difference-of-means direction is the normalised difference of class means of Δh (or of h for state tasks). Bootstrap intervals resample test questions (1,000 draws) using the stored probe weights.

Nuisance covariates. Δ tokens, total tokens, Δ gold log-probability, Δ next-token entropy, Δ top probability, answer string in added text / in the full context / in the previous context, TF-IDF cosine relevance of the added passage to the question, number and word count of added paragraphs, relative insertion position, and a 16-dimensional truncated-SVD embedding of the added text’s TF-IDF vector (fit on training increments). The residualised probe regresses Δh on these with ridge ($\lambda = 1$) and probes the residual. The “gain” adds out-of-fold (5-fold, grouped by question) probe scores to the nuisance classifier.

Text baselines. TF-IDF (1–2-grams, sublinear tf, $\leq 200k$ features) with logistic regression; DeBERTa-v3-base fine-tuned with AdamW (lr 3×10^{-5} , linear decay, class-weighted cross-entropy). The reduced budget uses one epoch, 256 tokens and 8,000 training increments (all positives kept); the full budget two epochs, 512 tokens and all training increments.

Interventions. Steering adds the vector at the anchor and every generated position (“anchor onward”) or at all positions; the unit u is $|\mathbb{E}[v^\top h \mid \text{sufficient}] - \mathbb{E}[v^\top h \mid \text{insufficient}]|$ on training conditions, computed per layer. Projection-out sets $v^\top h$ to the insufficient-state mean at the anchor and generated positions for every layer from the best layer onward. Patching replaces the anchor residual during the prefill at every layer from the best layer to the last; decoding then proceeds from the patched cache. Principal components are computed from 1,500 training decisive updates per layer; the “unembedding” direction is the normalised row of the output embedding matrix for the gold answer’s first token.

C Additional results

Condition (correct / abstain)	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
no evidence	0.00 / 1.00	0.00 / 1.00	0.00 / 0.99
distractors only	0.00 / 0.97	0.00 / 0.97	0.00 / 0.94
one hop	0.10 / 0.80	0.12 / 0.76	0.13 / 0.69
minimal sufficient	0.46 / 0.30	0.47 / 0.31	0.36 / 0.36
one hop + false final hop	0.13 / 0.49	0.13 / 0.55	0.12 / 0.51
sufficient + distractor	0.39 / 0.35	0.38 / 0.41	0.27 / 0.47
full context (20 par.)	0.22 / 0.38	0.17 / 0.61	0.17 / 0.48

Table 5: Behaviour by condition family (evidence-only regime, 2,000 questions per model): fraction correct / fraction answering INSUFFICIENT.

Run	Task	question	document	answer	relation
Qwen2.5-7B	Suff.	0.956	0.948	0.945	0.947
Qwen2.5-7B	Uptake	0.851	0.775	0.726	0.841
Llama-3.1-8B	Suff.	0.960	0.952	0.958	0.959
Llama-3.1-8B	Uptake	0.845	0.783	0.736	0.815
Mistral-7B	Suff.	0.964	0.949	0.954	0.958
Mistral-7B	Uptake	0.873	0.771	0.758	0.793
Qwen 3–4 hop	Suff.	0.931	0.925	0.923	0.932
Qwen 3–4 hop	Uptake	0.795	0.613	0.611	0.795
Qwen 2Wiki	Suff.	0.986	0.986	0.981	0.983
Qwen 2Wiki	Uptake	0.894	0.894	0.866	0.786

Table 6: Best-layer AUROC under the four disjoint splits, for the three 2-hop runs, the 3–4-hop run and 2WikiMultiHopQA.

Model	α	sufficient corr./ans.	insufficient ans.	distractors ans.	false ans./followed	answer-string ans.
Qwen2.5-7B	-1.0	0.23/0.34	0.05	0.00	0.23/0.06	0.03
	-0.5	0.34/0.52	0.09	0.01	0.43/0.13	0.09
	+0.5	0.49/0.79	0.20	0.06	0.75/0.25	0.29
	+1.0	0.49/0.82	0.26	0.12	0.79/0.25	0.35
Llama-3.1-8B	-1.0	0.03/0.07	0.00	0.00	0.03/0.01	0.00
	-0.5	0.25/0.38	0.07	0.00	0.23/0.12	0.05
	+0.5	0.65/0.89	0.27	0.13	0.82/0.35	0.34
	+1.0	0.69/0.94	0.51	0.29	0.92/0.39	0.59
Mistral-7B	-1.0	0.01/0.08	0.06	0.11	0.06/0.01	0.04
	-0.5	0.11/0.33	0.09	0.01	0.25/0.09	0.04
	+0.5	0.47/0.78	0.31	0.12	0.75/0.29	0.29
	+1.0	0.54/0.87	0.47	0.28	0.87/0.28	0.43

Table 7: Low-dose steering along v_{dm} (150 held-out questions per set). Baselines ($\alpha = 0$) appear in Fig. 5.

Evidence-augmented regime. With the instruction “use the supplied evidence together with your own knowledge” the Qwen probes give sufficiency 0.945, matched 0.913, uptake 0.721, correction 0.786 (document split), within 0.05 of the evidence-only regime; the model answers more (sufficient: 0.51 correct, 0.23 abstain) and follows false evidence more.

Logistic-direction steering. Steering along the logistic state-sufficiency direction at the best layer, anchor onward, at $\alpha \in \{\pm 2, \pm 1, \pm 0.5\}$ gap units ($\leq 1.4 \ell_2$) and at $\alpha \in \{\pm 8, \pm 16, 32\}$ (5–22 ℓ_2 on a mean residual norm of 61), and along the uptake direction at layer 26 ($\leq 16 \ell_2$), leaves every rate within ± 0.02 of baseline on all five evaluation sets, as does the random control; at 32 units all rates degrade by 0.02–0.05 for probe and random alike. An all-positions variant at $\leq 11 \ell_2$ is equally flat.

High-dose difference-of-means steering. Adding $\alpha u_l v_{\text{dm}}^{(l)}$ at all positions and all layers from the best layer onward with $|\alpha| \geq 2$ (per-layer gaps of 5–14 ℓ_2) produces degenerate output (every prompt “answered”, never correctly, never INSUFFICIENT) for the probe direction and the random control alike, which motivated the four-layer, anchor-onward, $|\alpha| \leq 1$ protocol of Fig. 5.

Single-layer patching. Patching only the best sufficiency layer (16 for Qwen) leaves the insufficient prompt unchanged in all variants; patching only layer 26 with the full sufficient residual restores 0.18/0.62, so late-layer anchor content suffices to answer.

3–4-hop questions. Sufficiency probes: 0.931/0.925/0.923/0.932 (question/document/answer/relation); matched 0.907/0.900/0.896/0.906; uptake 0.795/0.613/0.611/0.795. Mean projection on the sufficiency direction is +0.023 for both chain-decisive and leave-one-out-decisive increments, +0.003 for false, -0.02 distractor, -0.06 redundant and answer-string, -0.11 non-decisive gold. Patching: none 0.05/0.20, full 0.25/0.62, parallel 0.06/0.19, orthogonal 0.25/0.62, PC256 0.20/0.51, random-256 0.06/0.20; projection-out of v_{dm} leaves 0.03 answered.

2WikiMultiHopQA. Within-dataset probes (document split): sufficiency 0.986, matched 0.972, uptake 0.894, correction 0.914; here the matched task is almost fully explained by nuisance covariates (0.965, chiefly the gold log-probability change), residualised 0.82.

Closed-book transfer. On 4,000 closed-book activations (correct 1.5%, abstain 91.7%): state-correctness direction 0.810 (best layer 0.838), correction 0.799, state-sufficiency 0.722 (0.808), abstention flag 0.977 for abstention; own-probe ceiling 0.874 / 0.973.

Adaptive retrieval (Qwen). Probe policy 0.442 accuracy, 2.87 rounds, 381 prompt tokens; behavioural rule 0.460, 3.09, 402; entropy threshold 0.404, 3.92; oracle 0.475, 3.00; full context 0.404, 4.00.

Conversational protocol. Probes trained within the conversational regime reach sufficiency 0.951, uptake 0.818, correction 0.922, stability 0.959 and revision 0.969 (document split), the last two much higher than in the stateless regime because the model’s previous answer is in the prompt.

D Compute and reproducibility

Collection for one 2-hop model ($\approx 35k$ conditions) takes about 22 hours on 64 CPU cores (bf16, ≈ 300 prompt tokens/s) or about one hour on an H200 shared with another job; the 3–4-hop run took 1.3 GPU-hours; all causal experiments and text baselines together used about 7 GPU-hours. Probing runs on CPU in one to two hours per model. Every stage is deterministic given the seed (question sampling, insertion positions, false-answer choice, splits) and resumable. Code, configurations and the trajectory construction are released with the paper.